ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** stores the scarce rain for later use	Rain falls in short spells
2	**(b)** repairing leaking taps and reusing water	Reduces wastage
3	**(d)** drip and sprinkler irrigation	Water goes directly to roots
4	**(d)** factories use large amounts of water and pollute it	Industries compete with farms
5	**(b)** to irrigate crops and store water	Agricultural prosperity
6	**(b)** block the natural flow of the river	Fish cannot move upstream to spawn
7	**(b)** it is a renewable but limited resource	Fresh water is limited
8	**(d)** Tamil Nadu	A model for others
9	**(a)** makes water unfit for use	Industrial and domestic waste
10	**(a)** community water management	Villagers manage it

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water is carried by bamboo pipes (1); It reaches plants drop by drop with little loss (1).	1+1
12	They draw attention to displacement and ecological costs (1); They push for better planning and resettlement (1).	1+1
13	They use local materials and rain (1); They avoid displacement and damage to ecology (1).	1+1
14	Rivers flow through many states (1); A dam in one state reduces flow for others, leading to demands over shares (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The Sardar Sarovar Dam on the Narmada would displace many tribals (1); Activists claimed rehabilitation was poor (1); Supporters said it brings water and power to Gujarat (1).	1+1+1
16	Gendathur in Mysuru district has rooftop harvesting by households (1); About 200 houses have the system (1); Water meets domestic needs and recharges wells (1).	1+1+1
17	Industries can treat and reuse water (1); This reduces fresh water demand (1); Less waste is released into rivers (1).	1+1+1
	OR: Rulers built dams, lakes and canals (1); Chandragupta Maurya's time saw many irrigation works (1); Bhopal Lake and other tanks stored water (1).	1+1+1
18	Sedimentation is the deposit of silt in a reservoir (1); It reduces storage capacity (1); It lowers irrigation and power output and increases flood risk (1).	1+1+1
	OR: Large dams are costly, displace people and have wider ecological effects (1); Small structures are cheap, local and ecologically sound (1); Large dams give power and big irrigation, small ones recharge groundwater (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dams displace people and submerge land (1); Locals felt ignored in planning (1); Narmada Bachao Andolan opposed the Sardar Sarovar Dam (1); Protests also arose against Tehri dam (1); Movements demanded rehabilitation and review (1).	1+1+1+1+1
	OR: Policy can regulate use and promote harvesting (1); Tamil Nadu's law is an example (1); Technology like drip irrigation saves water (1); Recycling reduces demand (1); Both are needed together (1).	1+1+1+1+1
20	Rajasthan has tankas, johads and khadis (1); Meghalaya has bamboo drip irrigation (1); Himalayas have guls or kuls (1); Bengal had inundation channels (1); Karnataka's Gendathur and Shillong use rooftop harvesting (1).	1+1+1+1+1
	OR: Arid areas have little and erratic rain (1); Harvesting stores it for dry spells (1); It recharges groundwater (1); It is cheap and local (1); It reduces dependence on distant sources (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** multiple cropping with irrigation	Green Revolution areas
2	**(c)** Shillong	In Meghalaya
3	**(b)** johads	Traditional structure of Rajasthan
4	**(c)** rooftop harvesting	Householders harvest rooftop rain
5	**(a)** Hirakud	In Odisha
6	**(a)** great pressure on water resources	Per person availability is low
7	**(d)** Sardar Sarovar	Narmada Bachao Andolan
8	**(a)** seepage and evaporation	Unlined canals leak
9	**(d)** the flood plains of Bengal	Floodwater is diverted to fields
10	**(b)** serves more than one use such as irrigation, power and flood control	Integrates several uses

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Roof water is led into pits or wells (1); It percolates into the ground and raises the water table (1).	1+1
12	Water flow downstream is reduced or released suddenly (1); Silt supply to fields falls and floods may occur (1).	1+1
13	Silt reduces reservoir capacity and heavy rain forces release of water (1); Sudden release floods downstream areas (1).	1+1
14	Water is limited and demand is growing (1); Conservation ensures supply for future generations (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The government makes laws on water use and sharing (1); It builds projects and promotes harvesting (1); It can enforce rules, as in Tamil Nadu (1).	1+1+1
16	Tankas store roof rain in a dry area (1); Khadins keep soil moisture in farms (1); Both use scarce rain and need little cost (1).	1+1+1
17	Households in Shillong harvest rain from roofs (1); Water is collected and stored in tanks (1); It supplements the city's limited supply (1).	1+1+1
	OR: For: irrigation, power, flood control and water supply (1); Against: displacement, loss of forests and sedimentation (1); Critics say benefits are not equally shared (1).	1+1+1
18	Bengal flood plains had inundation channels (1); Floodwater was diverted to fields (1); It irrigated crops and enriched soil with silt (1).	1+1+1
	OR: Polluted water cannot be used for drinking or farming (1); Treatment costs more (1); Usable supply falls (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Benefits are irrigation, power, flood control and water supply (1); They support industry and agriculture (1); Costs are displacement and loss of forests (1); Sedimentation reduces capacity (1); Planning and resettlement are needed (1).	1+1+1+1+1
	OR: Rivers pass through several states (1); States compete for water (1); Krishna-Godavari is a dispute (1); Tribunals and negotiation are used (1); Sharing agreements and efficient use reduce conflict (1).	1+1+1+1+1
20	Forests hold rainwater and release it slowly (1); Roots increase percolation (1); Deforestation raises run-off and floods (1); It dries up streams (1); Protecting forests secures water (1).	1+1+1+1+1
	OR: Ancient rulers built dams, tanks and canals (1); Bhopal Lake and Hauz Khas tank are examples (1); Modern times use big dams and tube wells (1); Modern methods supply more but harm ecology (1); A mix of both can be sustainable (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** growing population and pollution	Demand rises and sources are polluted
2	**(a)** Hirakud	It also supplies irrigation and power
3	**(a)** Sringaverapura	Ganga floodwater system
4	**(a)** over-extraction through tube wells	Extraction exceeds recharge
5	**(a)** sharing of river water	Examples: Krishna-Godavari
6	**(c)** reduces the storage capacity of the reservoir	Silt builds up
7	**(a)** most of it is locked as ice or lies deep underground	Surface water and groundwater are small shares
8	**(c)** loss of forests and change in river ecosystems	Submergence of land
9	**(d)** guls or kuls	Channels from streams
10	**(a)** displacement of people	Villages are submerged

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A tanka is dug in the house and connected to the roof by pipes (1); It stores rain for the dry months (1).	1+1
12	Water is drawn faster than rain refills it (1); The water table falls and wells dry up (1).	1+1
13	It is renewed by the water cycle (1); But the usable fresh amount is small and unevenly spread (1).	1+1
14	Kuls carry stream water to fields on hill slopes (1); This lets farmers irrigate terraced fields without big dams (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Too much rain in a place causes floods (1); Lack of rain causes drought (1); Both need planning to store and manage water (1).	1+1+1
16	High-yielding crops need assured irrigation (1); Farmers sank tube wells and pumped groundwater (1); Water tables fell in Punjab, Haryana and western UP (1).	1+1+1
17	Rainfall is uneven in space and time (1); Large population and heavy demand strain supply (1); Pollution and over-extraction reduce usable water (1).	1+1+1
	OR: Governments regulate tube wells and licences (1); Some states promote recharge and harvesting (1); Laws and awareness help prevent over-extraction (1).	1+1+1
18	Communities build and maintain johads and kuls (1); They make rules for equal sharing (1); Their involvement makes systems durable (1).	1+1+1
	OR: Repair leaking taps (1); Reuse water for plants (1); Harvest rainwater (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Local communities know their needs (1); Local structures are cheap (1); They avoid displacement (1); People feel ownership (1); Decentralised systems are more sustainable (1).	1+1+1+1+1
	OR: Water is shared by many users (1); Scarcity causes disputes between states and users (1); Cooperation through tribunals and agreements solves them (1); Community management shows cooperation (1); Fair sharing is the key (1).	1+1+1+1+1
20	Population growth raises water demand (1); Irrigation takes most water (1); Industry and cities add demand and pollution (1); Groundwater is over-used (1); Effects are dry wells, conflicts, poor crops and health problems (1).	1+1+1+1+1
	OR: Irrigation uses large quantities of groundwater (1); Industry uses and pollutes water (1); Cities pave land and reduce recharge (1); Together they lower the water table (1); Recharge and regulation are required (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND